

Warm-up 10

10.2 For each matrix, write associated matrix A and find matrix $H \in SO(2)$ which diagonalizes A

a) $\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}$

→ quadratic eq. with associated matrix: $6x^2 + 9y^2 + 4xy + 42 = 0$
random ct :)

H = the base change matrix $H e' e$

$e \rightarrow$ basis w.r. to which the matrix A is given.

$e' \rightarrow$ orthonormal basis of eigenvectors.

$$\boxed{\det(A - \lambda I_2)} = \begin{vmatrix} 6-\lambda & 2 \\ 2 & 9-\lambda \end{vmatrix} = (6-\lambda)(9-\lambda) - 4 = 0$$

$$\Rightarrow 54 - 15\lambda + \lambda^2 - 4 = 0$$

$$\Rightarrow \lambda^2 - 15\lambda - 50 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 5) = 0. \quad \begin{matrix} \rightarrow \lambda_1 = 10 \\ \rightarrow \lambda_2 = 5 \end{matrix}$$

$$\boxed{Av = \lambda v}$$

$$\bullet \lambda_2 = 5 \Rightarrow A \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 5 \cdot \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow x \cancel{= 1}$$

$$\Rightarrow (A - \lambda I_2)v = 0 \Rightarrow (A - 5I_2) \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow$$

$$\Rightarrow \left(\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{cases} x + 2y = 0 \\ 2x + 4y = 0 \end{cases} \quad | :2 \Rightarrow \begin{cases} x + 2y = 0 \\ 2x + 4y = 0 \end{cases}$$

$$\Rightarrow x = -2y \Rightarrow v_2 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2y \\ y \end{bmatrix} = y \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\text{Eigenspace for } \lambda_2 = 5 \text{ is: } \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\} = \left\langle \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\rangle$$

$$\bullet \lambda_1 = 10 \Rightarrow (A - 10I_2)v = 0 \Rightarrow \left(\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix} - \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow$$

$$\Rightarrow \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{cases} -4x + 2y = 0 \\ 2x - y = 0 \end{cases} \quad | :2 \Rightarrow \begin{cases} -2x + y = 0 \\ x - \frac{1}{2}y = 0 \end{cases} \Rightarrow y = 2x$$

$$\Rightarrow v_1 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 2x \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Eigenspace for } \lambda_1 = 10 \text{ is: } \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} : t \in \mathbb{R} \right\} = \left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle$$

Orthonormal basis : $\rightarrow$ normalize vectors.

$$|v_2| = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$$

$$|v_1| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\hookrightarrow \left\{ \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\} \Rightarrow H \text{ could be } H_{ee'} = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$K \cdot K^T = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I_2.$$

$$\det K = \frac{1}{\sqrt{5}^2} (-4-1) = \frac{-5}{5} = -1 \Rightarrow K \in O(2)$$

We want $\det K = 1 \Rightarrow K \in SO(2)$.

$$\Rightarrow \text{replace } \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix} \text{ by } \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow K = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \Rightarrow \det K = \frac{1}{5} \cdot (4+1) = 1 \Rightarrow K \in SO(2).$$

$$K_{\text{new}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \Rightarrow K \cdot K^T = \dots = I_2 \checkmark$$

Moreover, $K^{-1}AK = K^TAK = \dots$

10.3. $x^2 - 4xy + y^2 - 6x + 2y + 1 = 0$

a) isometric can. eg.

$$Q = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \text{ associated matrix.}$$

$$\begin{vmatrix} 1-\lambda & -2 \\ -2 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - 4 = 0 \Rightarrow (1-\lambda-2)(1-\lambda+2) = 0 \Rightarrow$$

$$\Rightarrow (-1-\lambda)(3-\lambda) = 0 \Rightarrow \lambda_1 = -1 \Rightarrow \lambda_2 = 3.$$

$$\bullet \lambda_1 = -1 \Rightarrow Qv = \lambda v$$

$$\begin{bmatrix} 1+1 & -2 \\ -2 & 1+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Leftrightarrow \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Leftrightarrow$$

$$(2) \begin{cases} -2y = 0 \\ -2x = 0 \end{cases} \Rightarrow x = y = 0 \Leftrightarrow \begin{cases} 2x - 2y = 0 \\ -2x + 2y = 0 \end{cases} \Rightarrow x = y \Rightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\bullet \lambda_2 = 3 \Rightarrow \begin{bmatrix} 1-3 & -2 \\ -2 & 1-3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} -2x - 2y = 0 \\ -2x - 2y = 0 \end{cases} \Rightarrow \begin{cases} -2x = 2y \\ -x = y \end{cases} \Rightarrow \begin{bmatrix} x \\ -x \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Normalize vectors: $|v_1| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$$|v_2| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Orthormal basis: $B' = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right)$

Base change matrix: $M_{BB'} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$\det M = \frac{1}{2} (-1-1) = -\frac{2}{2} = -1 \quad \text{X. NOPE.}$$

$$M_{BB'} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \Rightarrow \det M = \frac{1}{2} (1+1) = \frac{2}{2} = 1 \checkmark$$

$$\hookrightarrow \text{a rotation by } \theta = -45^\circ \Rightarrow \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\Rightarrow \theta = -\frac{\pi}{4}$$

Eq. of our quadratic curve:

$$[x' y'] H^T Q H \begin{bmatrix} x' \\ y' \end{bmatrix} + [-6 \ 2] H \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0.$$

$$[x' y'] \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + [-6 \ 2] \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0.$$

$$[x' y'] \cdot \frac{1}{2} \begin{bmatrix} 3 & -3 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \frac{1}{\sqrt{2}} [-8 \ -4] \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0.$$

$$[x' y'] \cdot \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \dots = 0$$

$$[x' y'] \cdot \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \frac{1}{\sqrt{2}} [-8 \ -4] \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0$$

$$\begin{array}{r} 2 \ 1 \ 2 \ 2 \\ 1 \ 2 \ 2 \ 2 \end{array}$$

$$[3x' - y'] \begin{bmatrix} x' \\ y' \end{bmatrix} + \left[\frac{-8}{\sqrt{2}} \ \frac{-4}{\sqrt{2}} \right] \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0$$

$$3' \cdot \frac{16^4}{8 \cdot 4} \begin{array}{r} 1 \ 2 \ 2 \ 1 \\ 3 \end{array}$$

$$3x'^2 - y'^2 + -\frac{8}{\sqrt{2}}x' - \frac{4}{\sqrt{2}}y' + 1 = 0$$

$$3(x'^2 - 2 \cdot \frac{4}{3\sqrt{2}}x' + (\frac{4}{3\sqrt{2}})^2) - 3 \cdot (\frac{4}{3\sqrt{2}})^2 - (y'^2 + 2 \cdot \frac{2}{\sqrt{2}}y' + (\frac{2}{\sqrt{2}})^2) + (\frac{2}{\sqrt{2}})^2 + 1 = 0$$

$$3 \left(x' - \frac{4}{3\sqrt{2}} \right)^2 - \left(y' + \frac{2}{\sqrt{2}} \right)^2 - \frac{8}{3} + 2 + 1 = 0$$

$\underbrace{\hspace{2cm}}_{x''} \quad \underbrace{\hspace{2cm}}_{y''} \quad \underbrace{\hspace{2cm}}_{+\frac{19}{9}}$

$$(2) \ 3x''^2 - y''^2 + \frac{19}{9} = 0$$

$$(3x'' = x''')^2 \Rightarrow 3x''^2 = x'''^2$$

$$- (3y'' = -y''')^2 \Rightarrow 3y''^2 = -y'''^2$$

$$x'''^2 - 3y'''^2 - \frac{1}{9} = 0$$

$$3x''^2 - y''^2 = -\frac{1}{9} \quad | \cdot (-1)$$

$$-3x''^2 + y''^2 = \frac{1}{9} \quad | \cdot 9$$

$$-27x''^2 + 9y''^2 = 1 \quad | \cdot \frac{1}{27}$$

$$-\frac{x''^2}{\frac{1}{27}} + \frac{y''^2}{\frac{1}{9}} = 1 \quad \Rightarrow \quad \frac{x'''^2}{\frac{1}{9}} - \frac{y'''^2}{\frac{1}{27}} = 1. \quad (\text{hyp.})$$

$$x''' = y''$$

$$y''' = x'$$

b) affine eq. (Lagrange method) → affine change of coords.

$$x^2 - 4xy + y^2 - 6x + 2y + 1 = 0.$$

$$(a) (x^2 - 4xy + 4y^2) - 4y^2 + y^2 - 6x + 2y + 1 = 0$$

$$(e) (x - 2y)^2 - 3y^2 - 6x + 2y + 1 = 0$$

$$\underbrace{x - 2y}_{x_1}$$

$$y_1 = y$$

$$x - 2y = x_1$$

$$2) x = x_1 + 2y = x_1 + 2y_1$$

$$\left[\begin{matrix} x_1 \\ y_1 \end{matrix} \right] = \left[\begin{matrix} x - 2y \\ y \end{matrix} \right] = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \left[\begin{matrix} x \\ y \end{matrix} \right]$$

$$\Rightarrow x_1^2 - 3y_1^2 - 6(x_1 + 2y_1) + 2y_1 + 1 = 0$$

$$\Rightarrow x_1^2 - 3y_1^2 - 6x_1 - 10y_1 + 1 = 0$$

$$\Rightarrow (x_1 - 2 \cdot 3x_1 + 9) - 9 - (3y_1^2 + 2\sqrt{3}y_1 \frac{5}{\sqrt{3}} + \frac{25}{3}) + \frac{25}{3} + 1 = 0$$

$$\Rightarrow \underbrace{(x_1 - 3)^2}_{x_2} - \underbrace{(\sqrt{3}y_1 + \frac{5}{\sqrt{3}})^2}_{y_2} + \frac{1}{3} = 0$$

$$\left[\begin{matrix} x_2 \\ y_2 \end{matrix} \right] = \left[\begin{matrix} x_1 - 3 \\ \sqrt{3}y_1 + \frac{5}{\sqrt{3}} \end{matrix} \right] = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix} \left[\begin{matrix} x_1 \\ y_1 \end{matrix} \right] + \begin{bmatrix} -3 \\ \frac{5}{\sqrt{3}} \end{bmatrix}$$

$$\Rightarrow x_2^2 - y_2^2 = -\frac{1}{3} \quad | : (-\frac{1}{3})$$

$$\Rightarrow \underbrace{\left(\frac{y_2}{\frac{1}{\sqrt{3}}} \right)^2}_{x_3} - \underbrace{\left(\frac{x_2}{\frac{1}{\sqrt{3}}} \right)^2}_{y_3} = 1$$

$$\Rightarrow x_3^2 - y_3^2 = 1 \quad (\text{hyp.})$$

$$\frac{y_2}{\frac{1}{\sqrt{3}}} = x_3 \Rightarrow *$$

$$\Rightarrow \left[\begin{matrix} x_3 \\ y_3 \end{matrix} \right] = \left[\begin{matrix} \frac{y_2}{\frac{1}{\sqrt{3}}} \\ \frac{x_2}{\frac{1}{\sqrt{3}}} \end{matrix} \right]$$

$$= \begin{bmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{bmatrix} \left[\begin{matrix} x_2 \\ y_2 \end{matrix} \right] + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{matrix} x_3 \\ y_3 \end{matrix} \right] = \begin{bmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{bmatrix} \left[\begin{matrix} x_2 \\ y_2 \end{matrix} \right]$$

$$= \begin{bmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix} \left[\begin{matrix} x_1 \\ y_1 \end{matrix} \right] + \begin{bmatrix} -3 \\ \frac{5}{\sqrt{3}} \end{bmatrix} \right)$$

$$= \begin{bmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix} \left(\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \left[\begin{matrix} x \\ y \end{matrix} \right] \right) + \begin{bmatrix} -3 \\ \frac{5}{\sqrt{3}} \end{bmatrix} \right)$$

$$= \sqrt{3} \begin{bmatrix} 0 & \sqrt{3} \\ 1 & -2 \end{bmatrix} \left[\begin{matrix} x \\ y \end{matrix} \right] + \begin{bmatrix} 5 \\ -3\sqrt{3} \end{bmatrix}$$

→ the affine change of coords.